

AI-POWERED DOCUMENT SUMMARIZATION SYSTEM

A

MAJOR PROJECT-II REPORT

Submitted in partial fulfillment of the requirement.

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By

GROUP NO.42

Deepak Kurmi	0187CS221073
Ajay Ahirwar	0187CS221025
Alok Singh	0187CS221034
Anmol Banker	0187CS221049

Under the guidance of

Dr. Shweta Singh

(Associate Professor)



**Department of Computer Science & Engineering Sagar
Institute of Science & Technology (SISTec), Bhopal (M.P)**

**Approved by AICTE, New Delhi & Govt. of M.P.
Affiliated to Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal (M.P.)**

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Sagar Institute of Science & Technology (SISTec), Bhopal (M.P)

Department of Computer Science & Engineering



CERTIFICATE

We hereby certify that the work which is being presented in the B.Tech. Major Project-II Report entitled **“AI-Powered Document Summarization System”** in partial fulfillment of the requirements for the award of the degree of *Bachelor of Technology*, submitted to the Department of **Computer Science & Engineering**, Sagar Institute of Science & Technology (SISTec), Bhopal (M.P.) is an authentic record of our own work carried out during the period from Dec-2025 to May-2026 under the supervision of **Dr. Shweta Singh**.

The content presented in this project has not been submitted by me for the award of any other degree elsewhere.

Deepak Kurmi
0187CS221073

Ajay Ahirwar
0187CS221025

Alok Singh
0187CS221034

Anmol Bankar
0187CS221049

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Date:

Dr. Shweta Singh
Project Guide

Prof. Nargish Gupta
HOD

Dr. Manish Billore
Principal

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ABSTRACT

The AI-Powered Document Summarization System is an advanced tool designed to address the challenge of information overload. Using Natural Language Processing (NLP) and Machine Learning models, the system automatically condenses lengthy documents, such as research papers and reports, into concise and coherent summaries. Built with a robust backend and an intuitive user interface, the platform allows users to upload PDF or text files and receive instant summaries. This project aims to improve productivity for students and professionals by saving time spent on manual reading. It contributes to SDG 9 (Innovation) by leveraging AI to modernize information processing and administrative workflows. The AI-Powered Document Summarization System is an intelligent web-based application developed to solve the growing problem of information overload in the digital era. With the rapid increase in online documents, research papers, reports, articles, and organizational records, users often face difficulty in reading and extracting important information within limited time. Students, researchers, teachers, business professionals, and organizations spend a significant amount of time manually reviewing lengthy documents to identify key points and relevant insights. The proposed system addresses this challenge by utilizing Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning techniques to automatically generate concise, meaningful, and coherent summaries from large textual documents. The system is designed to improve productivity, reduce reading time, and enhance information accessibility through intelligent automation.

LIST OF ABBREVIATIONS

ACRONYM	FULL FORM
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma Separated Values
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JWT	JSON Web Token
NLP	Natural Language Processing
REST	Representational State Transfer
SDG	Sustainable Development Goals
UI	User Interface
UX	User Experience
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma Separated Values
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation

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CHAPTER 1

INTRODUCTION

1.1 ABOUT THE PROJECT

The AI-Powered Document Summarization System is an intelligent web-based application designed to simplify and automate the process of analyzing and summarizing lengthy documents. The project aims to reduce the time and effort required for reading large amounts of textual data by generating concise, meaningful, and accurate summaries using Artificial Intelligence and Natural Language Processing (NLP) techniques.

At its core, the system allows users to upload documents such as PDF, DOCX, or text files, from which the application extracts textual content and processes it using advanced AI-based summarization models. The generated summary highlights the most important information from the original document while maintaining the overall meaning and context.

The application is developed using modern technologies such as React.js for the frontend, Node.js and Express.js for backend services, and for secure data storage and management. The system provides a clean and user-friendly interface where users can upload documents, view generated summaries, manage uploaded files, and download summarized reports efficiently.

Additionally, the project includes features such as secure user authentication, summary history management, keyword extraction, document categorization, and downloadable summary reports. The platform can also support real-time processing and analytics to improve user productivity and accessibility of information.

1.2 PROPOSED SYSTEM

The proposed system, AI-Powered Document Summarization System, is designed to overcome the limitations of traditional document reading and manual information extraction by providing an intelligent, automated, and scalable web-based solution. The system uses Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to generate concise and meaningful summaries from lengthy documents, helping users save time and improve productivity.

Built using the MERN stack, the application provides users with a simple and efficient platform for uploading, processing, and managing documents. The system supports multiple document formats such as PDF, DOCX, and TXT files. Once a document is uploaded, the application extracts the textual content and processes it through AI-based summarization models to identify important sentences, keywords, and core concepts.

The frontend of the system is developed using React.js, offering a clean, responsive, and user-friendly interface for uploading files, viewing summaries, managing document history, and downloading summarized reports. The backend is developed using Node.js and Express.js, which handle document processing, API management, and communication between the frontend and the database. MongoDB is used for secure and structured storage of user information, uploaded documents, and generated summaries.

1.3 PROJECT OBJECTIVE

The main objective of the AI-Powered Document Summarization System is to develop a secure, intelligent, and efficient web application that automates the process of analyzing and summarizing lengthy documents. The project aims to reduce manual reading effort, improve information accessibility, and enhance productivity by using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques.

The system is designed to help students, researchers, professionals, and organizations quickly extract important information from large textual documents without reading the entire content manually. By integrating modern web technologies and AI-based summarization models, the project provides a fast, reliable, and user-friendly solution for efficient document management and text summarization.

Key objectives include:

- To develop an AI-based system capable of generating concise and meaningful summaries from lengthy documents.
- To automate the extraction and processing of textual content from documents such as PDF, DOCX, and TXT files.
- To implement Natural Language Processing (NLP) techniques for identifying important sentences, keywords, and core concepts from the uploaded documents.
- To provide a simple and interactive user interface for uploading documents, viewing summaries, and managing document history.

- To ensure secure user authentication and protected access to uploaded documents and generated summaries.
- To enable users to download summarized reports for future reference and documentation purposes.
- To improve productivity and save time by reducing the need for manual reading and information extraction.
- To build a scalable and efficient system using MERN stack technologies such as Express.js, React.js, and Node.js.
- To support accurate and fast document summarization for academic, professional, and organizational use cases.

1.4 SCOPE OF STUDY

The scope of the AI-Powered Document Summarization System encompasses the analysis, design, development, and implementation of an intelligent web-based platform capable of automatically summarizing lengthy textual documents using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques. The study focuses on developing an efficient and user-friendly system that helps students, researchers, professionals, and organizations quickly extract meaningful information from large documents.

The project covers the integration of modern web technologies and AI models to improve productivity, reduce manual reading effort, and enhance information management processes.

The major areas covered within the scope of this study include:

Document Upload and Processing:

Handling document uploads in multiple formats such as PDF, DOCX, and TXT files, extracting textual content, and preparing the data for summarization.

AI-Based Document Summarization:

Implementing Artificial ++Intelligence and Natural Language Processing (NLP) techniques to analyze document content and generate concise, meaningful, and accurate summaries.

Text Extraction and Preprocessing:

Performing text cleaning, tokenization, sentence extraction, and preprocessing operations to improve summarization quality and accuracy.

Keyword and Important Sentence Extraction:

Identifying important keywords, phrases, and key sentences from documents to help users understand the core concepts quickly.

User Authentication and Security:

Providing secure user login and authentication mechanisms to protect uploaded documents and user data.

Summary Management:

Allowing users to view, manage, save, and download generated summaries for future reference and documentation purposes.

User-Friendly Interface:

Developing an interactive and responsive frontend using React.js for smooth document upload, summary generation, and summary visualization.

Performance and Scalability:

Designing the system to efficiently handle multiple users and large documents while maintaining performance and reliability.

Productivity Enhancement:

Reducing the time required for reading lengthy documents and improving decision-making by providing quick access to important information.

CHAPTER 2

SOFTWARE & HARDWARE REQUIREMENTS

2 HARDWARE REQUIREMENTS

2.1 DEVELOPMENT COMPUTER

A computer with at least 8GB RAM, Intel i5/Ryzen 5 processor or higher, and a minimum of 256GB SSD storage is recommended for smooth development and execution of the AI-Powered Document Summarization System.

2.2 SERVER MACHINE

A dedicated or cloud-based server is required for hosting the web application, backend APIs, database services, and AI-based summarization models. The recommended configuration includes a minimum of 4GB RAM, multi-core processor, and sufficient storage capacity for storing uploaded documents, summaries, and user data.

2.3 STABLE INTERNET CONNECTION

A stable internet connection is necessary for downloading libraries and dependencies, accessing cloud services, integrating AI/NLP APIs or models, database connectivity, and deploying the application to hosting platforms.

2.4 DEBUGGING TOOLS

Development and debugging tools such as browsers with developer tools, terminal access, and code editors are required for testing, troubleshooting, and monitoring the application during development and deployment.

2.5 BACKUP DEVICES

External or cloud-based backup solutions are recommended for maintaining secure backups of source code, project documentation, uploaded files.

USB drives or external hard disks may be used for transferring project files, deployment builds, datasets, AI models, and backup copies between systems when required.

2.6 POWER BACKUP

An Uninterrupted Power Supply (UPS) is recommended to protect the development and server systems from unexpected power failures, preventing data corruption, loss of unsaved work, and interruptions during document processing or model execution.

SOFTWARE REQUIREMENTS

2.1.1 OPERATING SYSTEM

Windows, macOS, or Linux operating systems with support for modern web development technologies such as Node.js, MongoDB, React.js, and AI/NLP libraries required for document summarization.

2.1.2 INTEGRATED DEVELOPMENT ENVIRONMENT (IDE)

Visual Studio Code (VS Code), WebStorm, or any modern IDE for developing and managing the frontend, backend, database integration, and AI-based document summarization modules.

2.1.3 PROGRAMMING LANGUAGES

JavaScript for frontend and backend development, React.js for user interface development, Node.js and Express.js for server-side processing, and Python for implementing Artificial Intelligence and Natural Language Processing (NLP) models used in document summarization.

2.1.4 FRONTEND TECHNOLOGIES

React.js, HTML5, CSS3, and Bootstrap or Tailwind CSS for designing a responsive, interactive, and user-friendly web interface.

2.1.5 BACKEND FRAMEWORK

Node.js and Express.js for handling API requests, document processing, authentication, and communication between frontend and database services.

2.1.6 AI/NLP LIBRARIES AND TOOLS

Python libraries and frameworks such as NLTK, SpaCy, Transformers, Hugging Face, or TensorFlow/PyTorch for implementing AI-powered text summarization and keyword extraction functionalities.

2.1.7 VERSION CONTROL SYSTEM

Git and GitHub for source code management, project collaboration, version tracking, and backup purposes.

2.1.8 WEB BROWSER

Google Chrome, Mozilla Firefox, or Microsoft Edge for testing and accessing the web application efficiently.

CHAPTER 3

PROBLEM DESCRIPTION

INTRODUCTION

In today's digital world, students, researchers, professionals, and organizations deal with a huge amount of textual information such as research papers, reports, articles, business documents, and academic materials. Reading and understanding these lengthy documents manually is a time-consuming and difficult task. Users often spend a significant amount of time identifying important information, extracting key points, and preparing summaries from large documents.

Traditional methods of document analysis and summarization are inefficient, repetitive, and prone to human error. Manually reading large volumes of content not only reduces productivity but also increases the possibility of missing critical information. In many cases, users struggle to quickly access concise and meaningful insights from documents, especially when handling multiple files or large datasets.

Additionally, most existing systems do not provide intelligent automation, accurate summarization, keyword extraction, or efficient document management within a single platform. The lack of a centralized and AI-driven solution results in inefficient information processing, poor accessibility, and increased workload for users.

Therefore, there is a need for a smart, automated, and secure platform that can efficiently process documents, extract important information, and generate meaningful summaries in less time. The AI-Powered Document Summarization System aims to address these challenges by using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to automate the summarization process and improve productivity.

3.1 ANALYSIS OF CURRENT PRACTICES

Currently, most users manually read and analyze lengthy documents to extract important information and create summaries. This process is slow, repetitive, and inefficient, especially when dealing with research papers, reports, articles, or large textual datasets. Users often need to spend hours reading documents to understand the core content, which negatively impacts productivity and decision-making efficiency.

Many existing document management systems only provide storage and viewing capabilities without intelligent summarization features. Some available summarization tools lack accuracy, context understanding, support for multiple file formats, or secure document management. Additionally, users face difficulties in organizing documents, managing summaries, and retrieving important information quickly.

The absence of an integrated AI-based system for automatic summarization, keyword extraction, and document management creates challenges such as increased manual workload, information overload, poor accessibility, and inefficient knowledge extraction.

The AI-Powered Document Summarization System addresses these issues by providing a centralized, intelligent, and automated platform capable of processing documents, extracting meaningful insights, generating concise summaries, and improving overall efficiency. The system helps users save time, reduce.

3.2 PROBLEM IDENTIFICATION

The scope of this project includes designing and developing a web-based AI-powered system that automates the process of document analysis and summarization. The project focuses on creating an intelligent platform capable of processing lengthy textual documents and generating concise, meaningful, and accurate summaries using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques.

The system covers uploading and processing multiple document formats such as PDF, DOCX, and TXT files, extracting textual content, identifying important sentences and keywords, and generating summarized output automatically. The project also includes implementing document management features that allow users to store, view, manage, and download generated summaries efficiently.

Additionally, the system provides secure user authentication and database integration to ensure safe handling of uploaded documents and user information. A responsive and user-friendly interface is developed to simplify document upload, summary generation, and summary visualization processes

3.3 PROPOSED SOLUTION

The AI-Powered Document Summarization System is a modern web-based application developed to simplify and automate the process of analyzing and summarizing lengthy textual documents.

The system uses Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to extract important information and generate concise, meaningful summaries from documents such as PDFs, DOCX files, and text documents.

The application provides users with an easy-to-use platform for uploading documents, processing textual content, viewing generated summaries, and managing summarized records efficiently. Developed using the MERN stack, the system combines React.js for the frontend interface, Node.js and Express.js for backend processing, and MongoDB for secure data storage and management.

The project includes features such as automatic text extraction, AI-based summarization, keyword extraction, summary history management, downloadable summarized reports, and secure user authentication. The system helps users reduce manual reading effort, save time, and improve productivity by providing quick access to the most important information from large documents.

With its intelligent automation, user-friendly interface, and scalable architecture, the AI-Powered Document Summarization System offers a reliable and efficient solution for students, researchers, professionals, and organizations. Overall, the project supports digital transformation by improving information accessibility, reducing workload, and enabling smarter document management through AI-driven technologies.

3.3 PROPOSED SOLUTION

The AI-Powered Document Summarization System is a modern web-based application developed to simplify and automate the process of analyzing and summarizing lengthy textual documents. The system uses Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to extract important information and generate concise, meaningful summaries from documents such as PDFs, DOCX files, and text documents.

The application provides users with an easy-to-use platform for uploading documents, processing textual content, viewing generated summaries, and managing summarized records efficiently. Developed using the MERN stack, the system combines React.js for the frontend interface, Node.js and Express.js for backend processing, and MongoDB for secure data storage and management.

The project includes features such as automatic text extraction, AI-based summarization, keyword extraction, summary history management, downloadable summarized reports, and secure user authentication. The system helps users reduce manual reading effort, save time, and improve productivity by providing quick access to the most important information from large documents.

CHAPTER 4

LITERATURE SURVEY

The study by Gupta and Verma (2022) focuses on the use of Artificial Intelligence and Natural Language Processing (NLP) techniques for automatic text summarization. Their research explains how AI models can analyze large textual documents, identify important information, and generate concise summaries efficiently. The study highlights that AI-based summarization systems significantly reduce manual reading effort, improve productivity, and enhance information accessibility for users dealing with large datasets. [1]

Sharma and Patel (2021) discuss the role of Natural Language Processing in intelligent document analysis systems. Their work emphasizes text preprocessing techniques such as tokenization, stop-word removal, sentence extraction, and keyword identification for improving summarization accuracy. The research demonstrates how NLP techniques can help in extracting meaningful insights from lengthy documents while maintaining contextual relevance. [2]

Singh and Arora (2023) explore the implementation of machine learning and deep learning models for extractive and abstractive text summarization. Their study compares various AI-based summarization approaches and highlights the effectiveness of transformer-based models such as BART and T5 in generating human-like summaries. They conclude that AI-powered summarization improves efficiency and provides accurate content understanding in academic and professional environments. [3]

The book *Natural Language Processing with Python* by Steven Bird, Ewan Klein, and Edward Loper provides detailed knowledge about NLP concepts, text processing, tokenization, parsing, sentiment analysis, and text summarization techniques. It serves as an important resource for understanding the implementation of NLP-based systems and language processing applications. [4]

The book *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* by Aurélien Géron explains the fundamentals of machine learning and deep learning algorithms used in intelligent systems. The book covers neural networks, text classification, sequence models, and transformer architectures useful for AI-powered document summarization applications. [5]

The official Hugging Face Transformers documentation provides comprehensive guidance on implementing pretrained transformer models such as BERT, GPT, BART, and T5 for Natural Language Processing tasks. It explains model training, fine-tuning, tokenization, and inference processes for text summarization systems. The documentation is highly valuable for developers building AI-based summarization applications. [6]

The official MongoDB documentation provides detailed information about NoSQL database management, schema design, CRUD operations, indexing, aggregation, and cloud deployment using MongoDB Atlas. It helps developers build scalable and secure database systems for storing uploaded documents, summaries, and user information in full-stack applications. [7]

The official React documentation provides a modern and detailed guide for building responsive and interactive user interfaces using React.js. It explains concepts such as components, hooks, state management, routing, and API integration that are essential for developing user-friendly web applications like document summarization systems. [8]

The Express.js documentation explains backend routing, middleware handling, RESTful API development, and server-side architecture using Node.js and Express.js. It helps developers create scalable backend services for document upload, processing, authentication, and communication between frontend and database systems. [9]

The official Node.js documentation provides an in-depth understanding of asynchronous programming, event-driven architecture, server development, and API management. It serves as a foundational resource for building high-performance backend systems capable of handling document processing and AI integration efficiently. [10]

CHAPTER 5

SOFTWARE REQUIREMENT SPECIFICATION

5.1 FUNCTIONAL REQUIREMENTS

5.1.1 USER REGISTRATION AND LOGIN

Users should be able to register and log in securely using their email and password. Authentication and authorization should be implemented using JWT-based secure login mechanisms to protect user accounts and uploaded documents.

5.1.2 USER & ADMIN MANAGEMENT

Users can upload, view, manage, and delete their documents and generated summaries. Administrators can monitor system activities, manage user accounts, and maintain database records securely.

5.1.3 EXCEL DATA UPLOAD & EXTRACTION

Users can upload documents in formats such as PDF, DOCX, and TXT files. The system extracts textual content from uploaded documents automatically for further AI-based processing and summarization.

5.1.4 EMAIL AUTOMATION

The system should generate concise and meaningful summaries from lengthy documents using Artificial Intelligence and Natural Language Processing (NLP) techniques. The summarization process should identify important sentences, keywords, and core concepts accurately.

5.1.5 EMAIL TEMPLATE MANAGEMENT

The system should automatically extract important keywords and phrases from uploaded documents to help users quickly understand the main topics and concepts.

5.1.6 SUMMARY HISTORY MANAGEMENT

Users should be able to view previously generated summaries and access uploaded document history.

The system should store summary records securely in the database for future reference.

5.1.7 DOWNLOADABLE SUMMARY REPORTS

Users should be able to download generated summaries in formats such as PDF or text files for academic, professional, or documentation purposes.

5.2 NON-FUNCTIONAL REQUIREMENTS

5.2.1 PERFORMANCE

The system should process uploaded documents and generate summaries within a few seconds depending on document size and complexity. The application should efficiently handle multiple users and document processing requests simultaneously without significant performance degradation.

5.2.2 RELIABILITY

The system should provide stable and reliable performance with minimal downtime. Proper error handling mechanisms should be implemented to manage document upload failures, summarization errors, and server-side exceptions effectively.

5.2.3 MAINTAINABILITY

The application should follow a modular and organized folder structure for frontend, backend, and AI modules to simplify maintenance and future upgrades. The codebase and REST APIs should be properly documented to support easy debugging, enhancement, and scalability.

5.3 SECURITY

5.3.1 AUTHENTICATION AND AUTHORIZATION

JWT-based authentication should be implemented for secure user registration and login.

Role-based access control should be used to prevent unauthorized access to documents, summaries, and administrative functionalities.

CHAPTER 6

SOFTWARE DESIGN

6.1 USE CASE DIAGRAM

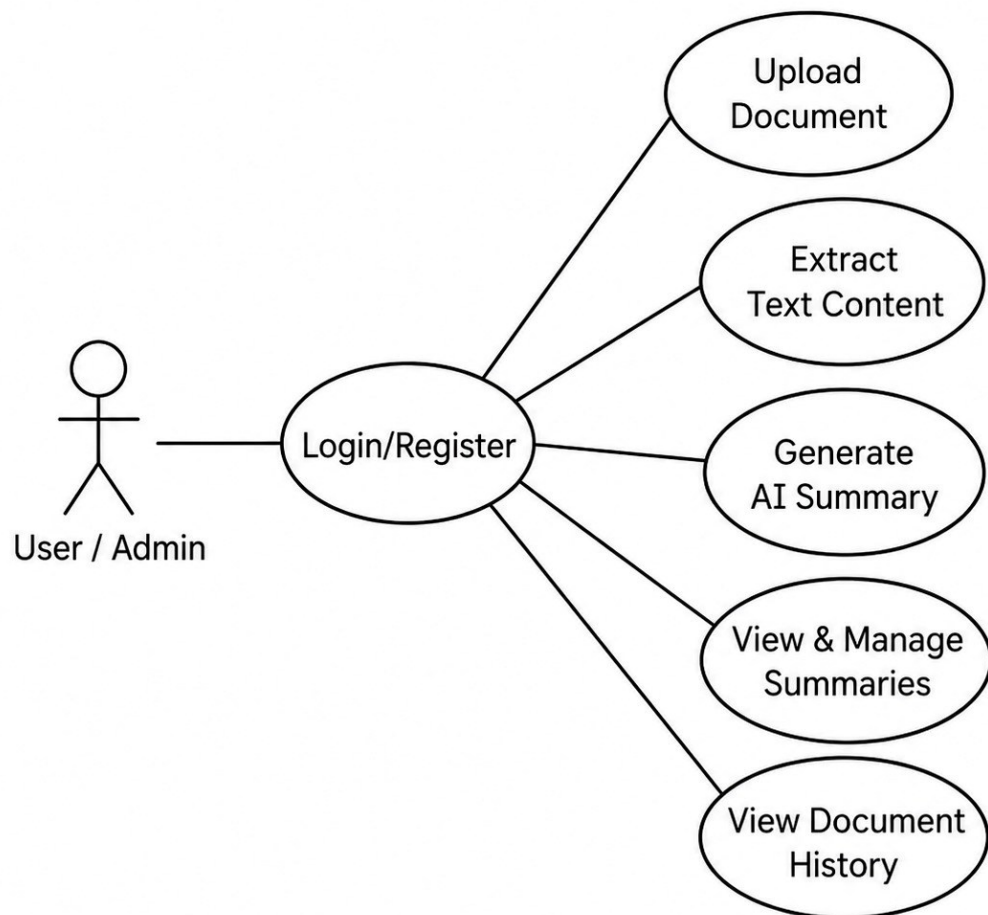


Figure 6.1: Use Case Diagram

6.2 ARCHITECTURE

6.2.1 WEB APPLICATION

Frontend: Developed using React.js for a responsive and user-friendly interface.

User Interaction: Allows administrators to upload Excel/CSV files, manage participants, create events, design certificates, and schedule automated emails.

UI/UX: Built with modern UI libraries ensuring accessibility, clarity, and smooth workflows.

Optimized for desktop and tablet devices used in academic or organizational environments.

6.2.2 BACKEND INFRASTRUCTURE

Framework: Node.js with Express.js for building RESTful APIs.

CHAPTER 7

WEB MODULE

7.1 INTRODUCTION

The Web Module of the AI-Powered Document Summarization System is designed to provide users with a seamless platform for uploading, processing, summarizing, and managing textual documents efficiently. Built using the MERN stack, the web module focuses on intelligent document analysis, AI-based summarization, keyword extraction, and efficient workflow handling. It supports document uploads in formats such as PDF, DOCX, and TXT, real-time processing updates, secure authentication, and summary management to ensure smooth and structured operations.

7.2 PROBLEM DEFINITION AND ALGORITHM

7.2.1 PROBLEM DEFINITION

Despite the availability of various document management and text processing systems, most existing platforms lack intelligent automation features such as AI-based document summarization, keyword extraction, real-time processing updates, and centralized document management. Users often rely on manual reading and analysis of lengthy documents, which is time-consuming, inefficient, and prone to missing important information.

The challenge lies in integrating document upload, text extraction, intelligent summarization, keyword identification, secure storage, and real-time processing into a single scalable system that reduces manual effort and improves productivity.

The Web Module aims to solve these challenges by:

- Automatically extracting text from uploaded PDF, DOCX, and TXT documents.
- Generating concise and meaningful summaries using AI and NLP techniques.
- Extracting important keywords and key phrases from documents.
- Providing real-time processing updates and notifications.
- Maintaining summary history and downloadable reports.
- Offering dashboards and logs for document management and analytics.
- Ensuring secure access through JWT-based authentication and authorization.

7.3 WEB MODULE DESCRIPTION

The Web Module is built using a modular MERN architecture that separates concerns across frontend UI, backend logic, database management, and AI-based document processing layers.

Frontend (React.js)

The frontend provides a dynamic, responsive, and user-friendly interface for document upload, summarization, and management. Key screens include:

- **Login Screen:** JWT-based secure authentication for authorized user access.
- **Dashboard:** Displays document statistics, summary history, analytics, and quick insights.
- **Document Upload Screen:** Allows users to upload PDF, DOCX, and TXT documents for processing.
- **Summary Viewer:** Displays AI-generated summaries, extracted keywords, and important sentences.
- **Document Management:** View, search, filter, and manage uploaded documents and generated summaries.
- **Report & Logs Screen:** Displays processing activities, system logs, errors, and performance metrics.

Backend (Java& python)

The backend serves as the core processing and automation engine of the AI-Powered Document Summarization System:

- Handles document upload, text extraction, and preprocessing from PDF, DOCX, and TXT files.
- Manages AI-based document summarization and keyword extraction processes using NLP techniques.
- Provides RESTful APIs for seamless frontend and backend communication.

7.4 RESULT ANALYTICS

The Web Module was evaluated for performance, usability, automation capability, and reliability using real-world datasets. Key observations include:

Fast Data Processing: Excel files of up to 10,000+ rows were processed efficiently.

Accurate Personalization: Emails and certificates were generated error-free using placeholder.

Real-time Alerts: Instant feedback through Socket.IO improved user experience during bulk.

High Usability Feedback: Test users reported intuitive navigation, clear dashboards, and easy .

CHAPTER 8

CODING

HOME SCREEN

```
//Home page
import React, { useState } from "react";
import FileUpload from "../components/FileUpload";
import SummaryBox from "../components/SummaryBox";

function App() {
  const [summary, setSummary] = useState(null);

  const appContainerStyle = {
    minHeight: "100vh",
    background: "linear-gradient(135deg, #f5f7fa 0%, #c3cfe2 100%)",
    padding: "60px 16px",
    fontFamily: 'Inter, -apple-system, system-ui, sans-serif',
    display: "flex",
    flexDirection: "column",
    alignItems: "center",
  };

  const headerStyle = {
    fontSize: "clamp(2.2rem, 5vw, 3rem)", // responsive font
    fontWeight: "900",
    background: "linear-gradient(90deg, #1e293b, #3b82f6)",
    WebkitBackgroundClip: "text",
    WebkitTextFillColor: "transparent",
    marginBottom: "10px",
    letterSpacing: "-0.03em",
    textAlign: "center"
  };

  const subheaderStyle = {
    color: "#475569",
    fontSize: "1rem",
    marginBottom: "40px",
    maxWidth: "520px",
    textAlign: "center",
    lineHeight: "1.6"
  };

  const mainCardStyle = {
    width: "100%",
    maxWidth: "750px",
    backgroundColor: "rgba(255, 255, 255, 0.85)",
    backdropFilter: "blur(12px)",
    padding: "32px",
    borderRadius: "20px",
  };
}
```

```

    boxShadow: "0 10px 30px rgba(0,0,0,0.08)",
    border: "1px solid rgba(255,255,255,0.5)",
    display: "flex",
    flexDirection: "column",
    gap: "24px",
    transition: "all 0.3s ease"
  };

  const sectionDivider = {
    borderTop: "1px solid #e2e8f0",
    paddingTop: "24px",
    animation: "slideUp 0.5s ease"
  };

  return (
    <div style={appContainerStyle}>
      <header>
        <h1 style={headerStyle}>Summarize.ai</h1>
        <p style={subheaderStyle}>
          Transform long documents into concise, AI-powered insights instantly.
        </p>
      </header>

      <main style={mainCardStyle}>
        { /* Upload Section */ }
        <div style={{ display: "flex", justifyContent: "center" }}>
          <FileUpload setSummary={setSummary} />
        </div>

        { /* Summary Section */ }
        {summary && (
          <div style={sectionDivider}>
            <SummaryBox summary={summary} />
          </div>
        )}
      </main>

      { /* Animation */ }
      <style>
        {`
          @keyframes slideUp {
            from {
              opacity: 0;
              transform: translateY(20px);
            }
            to {
              opacity: 1;
              transform: translateY(0);
            }
          }
        `}
      </style>
    </div>
  );
}

```

```

    boxShadow: "0 10px 30px rgba(0,0,0,0.08)",
    border: "1px solid rgba(255,255,255,0.5)",
    display: "flex",
    flexDirection: "column",
    gap: "24px",
    transition: "all 0.3s ease"
  };

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        { /* Upload Section */ }
        <div style={{ display: "flex", justifyContent: "center" }}>
          <FileUpload setSummary={setSummary} />
        </div>

        { /* Summary Section */ }
        {summary && (
          <div style={sectionDivider}>
            <SummaryBox summary={summary} />
          </div>
        )}
      </main>

      { /* Animation */ }
      <style>
        {`
          @keyframes slideUp {
            from {
              opacity: 0;
              transform: translateY(20px);
            }
            to {
              opacity: 1;
              transform: translateY(0);
            }
          }
        `}
      </style>
    </div>
  );
}

```

8.1 CREATE SOCIETY SCREEN CODE

```

bartModel.py
from transformers import pipeline

summarizer = pipeline("summarization", model="facebook/bart-large-cnn")

def generate_summary(text, length):

    # Length settings
    if length == "short":
        max_len = 80
        min_len = 30
    elif length == "medium":
        max_len = 150
        min_len = 60
    else:
        max_len = 220
        min_len = 100

    # Limit text
    if len(text) > 4000:
        text = text[:4000]

    # Chunking
    chunk_size = 600
    chunks = []

    for i in range(0, len(text), chunk_size):
        chunks.append(text[i:i + chunk_size])

    summaries = []

    for chunk in chunks:
        if len(chunk.strip()) < 50:
            continue

        try:
            result = summarizer(
                chunk,
                max_length=max_len,
                min_length=min_len,
                do_sample=False
            )
            summaries.append(result[0]['summary_text'])
        except Exception as e:
            print("Chunk error:", e)

    final_summary = " ".join(summaries)

# 📌 Bullet Points
points = final_summary.split(". ")
points = [p.strip() for p in points if len(p) > 20]

```

```

points = points[:5]

return {
  "summary": final_summary,
  "points": points
}

```

8.2 FILE UPLOAD CODE

```

import React, { useState } from "react";
import { uploadFile } from "../services/api";

const FileUpload = ({ setSummary }) => {
  const [file, setFile] = useState(null);
  const [length, setLength] = useState("medium");
  const [loading, setLoading] = useState(false);

  const handleUpload = async () => {
    if (!file) return alert("Please select a file");

    setLoading(true);

    try {
      const result = await uploadFile(file, length);

      setSummary(result); // 🔥 IMPORTANT CHANGE

    } catch (error) {
      console.error(error);
      alert("Error generating summary");
    }

    setLoading(false);
  };

```



```

return (
<div style={{
  width: "100%",
  maxWidth: "420px",
  margin: "auto",
  padding: "30px",
  borderRadius: "18px",
  background: "rgba(255,255,255,0.9)",
  backdropFilter: "blur(10px)",
  boxShadow: "0 20px 40px rgba(0,0,0,0.08)",
  border: "1px solid rgba(226,232,240,0.6)",
  display: "flex",
  flexDirection: "column",
  gap: "18px",
  textAlign: "center"
}}>

```

```

{/* Title */}
<h2 style={{
  margin: 0,
  fontSize: "1.3rem",
  fontWeight: "700",
  color: "#1e293b"
}}>

```

```

  Upload Document
</h2>

```

```

{/* Drag Area */}
<label style={{

```

```

border: "2px dashed #cbd5f5",
borderRadius: "14px",
padding: "30px",
cursor: "pointer",
transition: "all 0.25s ease",
background: "#f8fafc"
}}
onMouseOver={(e) => {
  e.target.style.borderColor = "#6366f1";
  e.target.style.background = "#eef2ff";
}}
onMouseOut={(e) => {
  e.target.style.borderColor = "#cbd5f5";
  e.target.style.background = "#f8fafc";
}}
>
<input
  type="file"
  onChange={(e) => setFile(e.target.files[0])}
  style={{ display: "none" }}
/>

<div style={{ fontSize: "0.9rem", color: "#64748b" }}>
  {file ? (
    <strong style={{ color: "#1e293b" }}>
      {file.name}
    </strong>
  ) : (
    "Drag & drop PDF here or click to upload"
  )}

```

```
</div>
```

```
</label>
```

```
{/* Dropdown */}
```

```
<div style={{ textAlign: "left" }}>
```

```
  <label style={{
    fontSize: "0.8rem",
    fontWeight: "600",
    color: "#64748b"
  }}>
```

```
    Summary Length
```

```
</label>
```

```
<select
```

```
  value={length}
```

```
  onChange={(e) => setLength(e.target.value)}
```

```
  style={{
    width: "100%",
    marginTop: "6px",
    padding: "10px",
    borderRadius: "10px",
    border: "1px solid #e2e8f0",
    background: "#fff",
    fontSize: "0.9rem",
    cursor: "pointer"
  }}
  >
```

```
<option value="short">Short</option>
```

```
<option value="medium">Medium</option>
```

```
<option value="long">Detailed</option>
```

```

</select>

</div>

{/* Button */}

<button

  onClick={handleUpload}

  disabled={loading}

  style={{

    background: loading

      ? "#94a3b8"

      : "linear-gradient(135deg, #6366f1, #3b82f6)",

    color: "#fff",

    padding: "12px",

    border: "none",

    borderRadius: "12px",

    fontWeight: "600",

    fontSize: "0.95rem",

    cursor: loading ? "not-allowed" : "pointer",

    boxShadow: loading

      ? "none"

      : "0 8px 20px rgba(99,102,241,0.35)",

    transition: "all 0.25s ease"

  }}

  onMouseOver={(e) => {

    if (!loading) {

      e.target.style.transform = "translateY(-2px)";

      e.target.style.boxShadow =

        "0 12px 28px rgba(99,102,241,0.45)";

    }

  }}

```

```

onMouseOut={e) => {
    e.target.style.transform = "translateY(0)";
    e.target.style.boxShadow =
        "0 8px 20px rgba(99,102,241,0.35)";
    }}
>
    {loading ? "Generating..." : "Generate Summary"}
</button>

</div>

);
};

export default FileUpload;

```

8.4. Summarizer.backend

```

package com.summarizer.backend.controller;

import org.springframework.web.bind.annotation.CrossOrigin;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.multipart.MultipartFile;

import com.summarizer.backend.model.SummaryResponse;
import com.summarizer.backend.service.SummarizationService;
import com.summarizer.backend.util.PdfExtractor;

```

```
@RestController
@CrossOrigin(origins = "*")
public class SummarizerController {

    private final SummarizationService service;

    public SummarizerController(SummarizationService service) {
        this.service = service;
    }

    @PostMapping("/summarize")
    public SummaryResponse summarize(
        @RequestParam("file") MultipartFile file,
        @RequestParam("length") String length) {

        String text = PdfExtractor.extractText(file);

        return service.getSummary(text, length);
    }
}
```

CHAPTER 9

RESULT AND OUTPUT SCREEN

9.1 RESULT

The development and implementation of the AI-Powered Document Summarization System successfully achieved all major project objectives. The system was designed and developed to simplify document analysis and automate the summarization process using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques. The application efficiently integrates React.js, HTML, CSS, and Bootstrap/Tailwind CSS for the frontend interface, Node.js and Express.js for backend processing, MongoDB for secure data management, and AI/NLP libraries such as Hugging Face Transformers, NLTK, BART, and T5 models for intelligent text summarization and keyword extraction.

AI-Based Document Summarization:

The system successfully generated concise, meaningful, and contextually accurate summaries from lengthy documents such as PDF, DOCX, and TXT files. AI models efficiently analyzed textual content and reduced large documents into understandable summarized information within seconds.

Fast Document Processing:

The application efficiently processed large textual documents with minimal delay. Uploaded files were parsed, analyzed, and summarized quickly, improving overall productivity and reducing manual reading effort.

Accurate Keyword Extraction:

The system accurately extracted important keywords, phrases, and key sentences from uploaded documents. This feature helped users quickly identify the core concepts and important topics within the document content.

Secure User Authentication:

JWT-based authentication ensured secure registration and login functionality. Users could securely access their uploaded documents, generated summaries, and history records without data inconsistency or unauthorized access.

Efficient Database Management:

MongoDB was used for securely storing user information, uploaded documents, generated summaries, keywords, and activity records. The database provided efficient read/write operations and smooth document management functionality.

Real-Time Processing Updates:

Integration of Socket.IO enabled real-time notifications and status updates during document upload and summary generation processes. Users received instant feedback regarding processing progress and completion status.

User-Friendly Interface:

The React.js-based frontend provided a responsive, interactive, and easy-to-use interface. Users were able to upload documents, generate summaries, manage records, and download summarized reports through a smooth and intuitive workflow.

Downloadable Summary Reports:

The system successfully allowed users to download summarized content in formats such as PDF or TXT for academic, professional, and documentation purposes.

Scalability and Performance:

The MERN-based architecture handled multiple user requests and document processing tasks efficiently without major performance issues. The system demonstrated scalability suitable for academic institutions, research environments, and professional organizations.

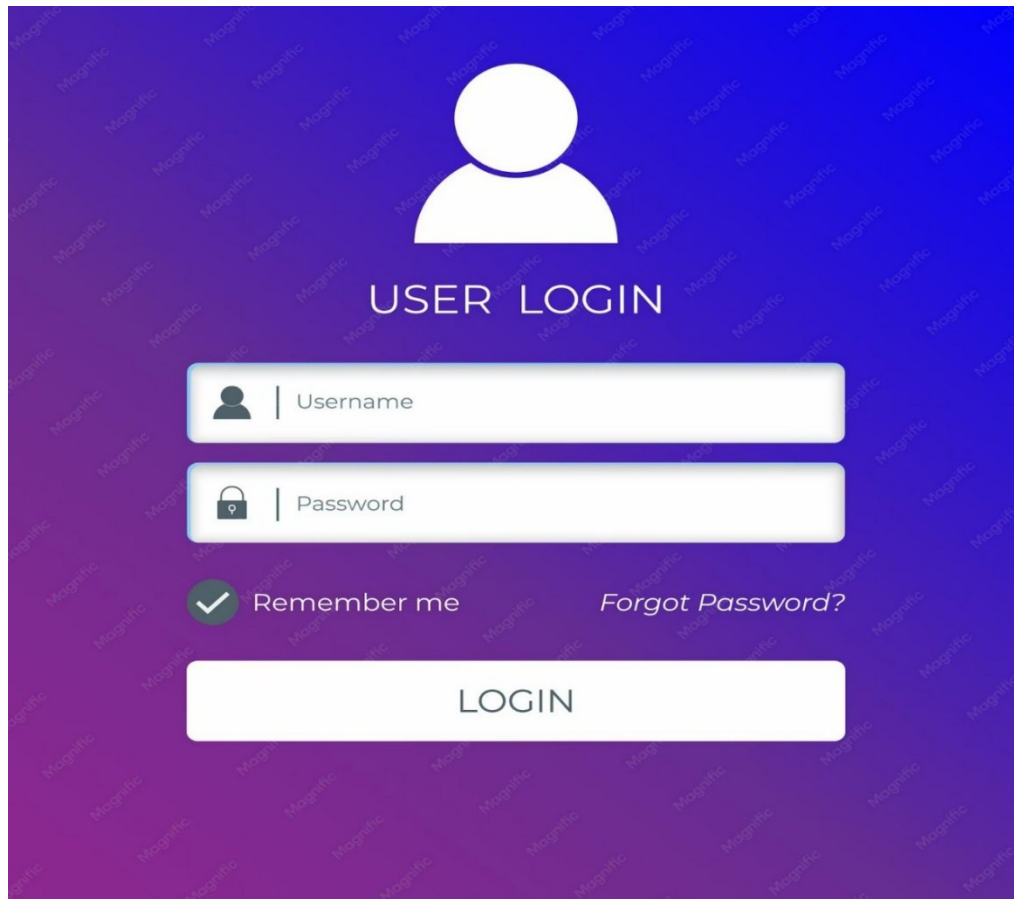
9.2 OUTPUT SCREENS

9.2.1 AUTHENTICATION SCREENS

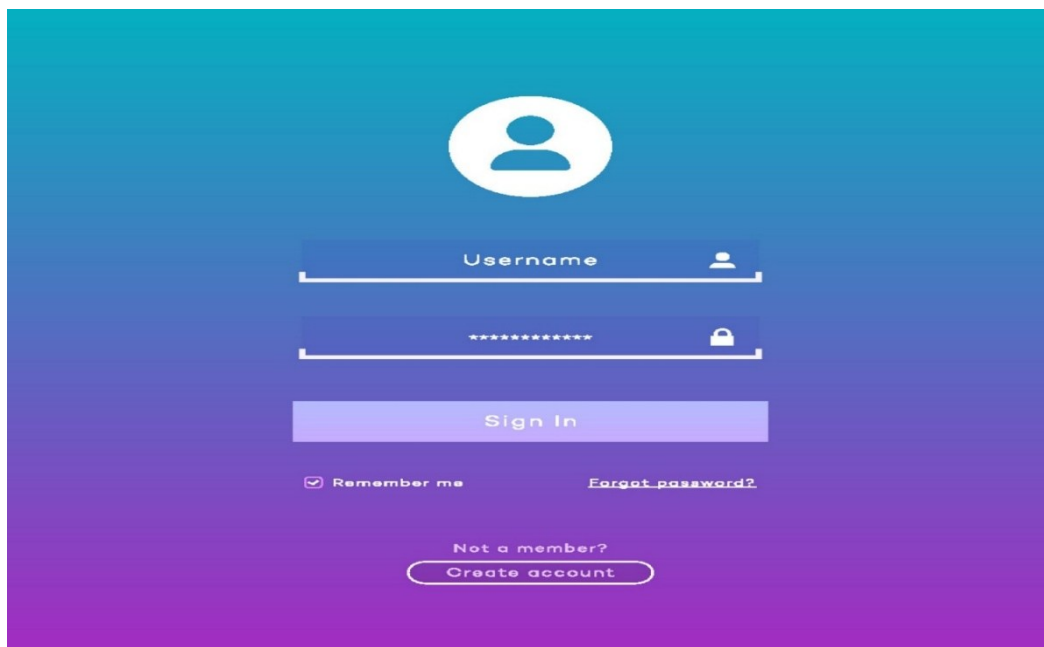
The authentication screen of the AI-Powered Document Summarization System provides a clean, secure, and user-friendly interface for users to access the application. Users can register and log in using their registered email and password credentials.

The authentication module includes the following features:

- Secure user registration and login functionality
- JWT-based authentication and authorization



9.1: Login Screen



9.2: Sign up Screen

9.2.2 DOCUMENT UPLOAD AND PROCESSING SCREENS

The document upload screen allows users to upload documents such as PDF, DOCX, and TXT files for AI-based summarization and text analysis. Users can easily select documents from their device and submit them through a simple and user-friendly interface.

The system automatically extracts textual content from the uploaded document and processes it using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques. During processing, users receive real-time status updates and notifications regarding upload progress and summary generation.

The screen also allows users to view document details, manage uploaded files, and access generated summaries efficiently.

Features of the Document Upload and Processing Screen:

- Upload PDF, DOCX, and TXT documents
- Automatic text extraction from uploaded files
- AI-based document summarization
- Real-time processing and status updates
- Keyword extraction and important sentence identification
- Secure document handling and storage

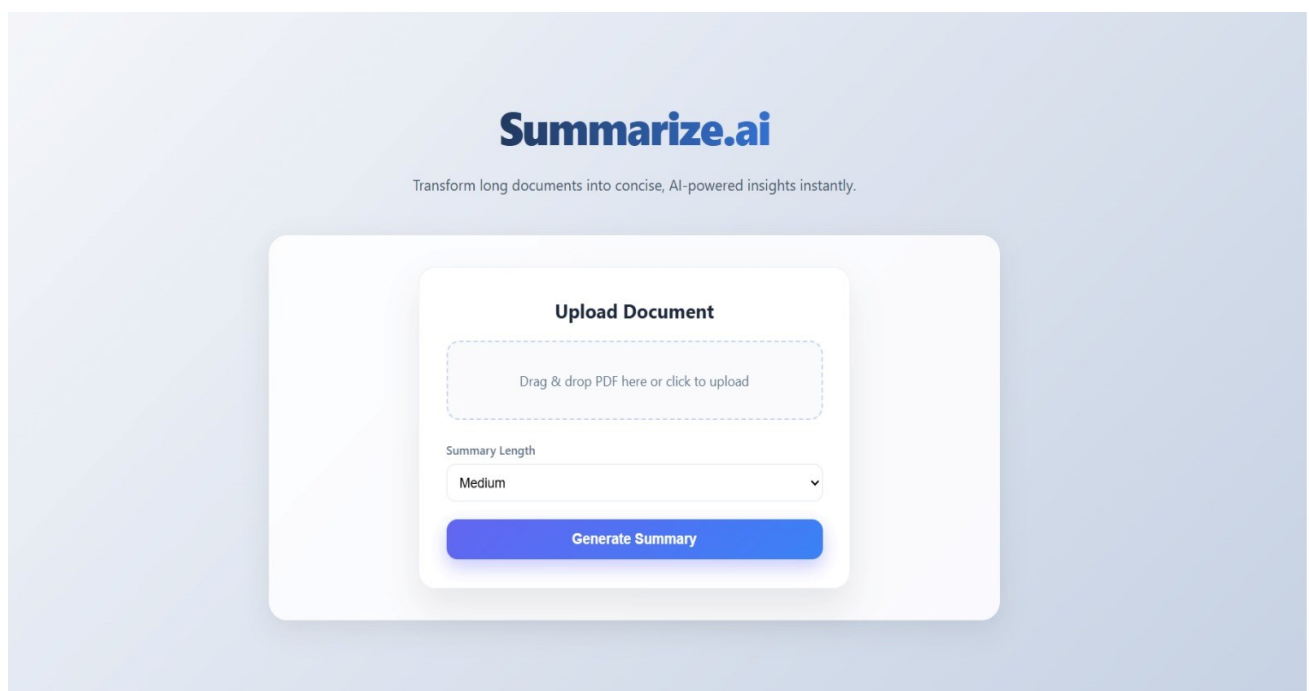


Figure 9.3 : The Document Upload and Processing Screen

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

The AI-Powered Document Summarization System successfully addresses the major challenges associated with manual document analysis and information extraction from lengthy textual content. By leveraging modern web technologies and Artificial Intelligence (AI) techniques, the system provides an intelligent, efficient, and user-friendly platform for automatic document summarization and keyword extraction.

The application successfully achieves all primary project objectives, including secure user authentication, document upload and processing, AI-based summarization, keyword extraction, real-time processing updates, downloadable summary reports, and efficient document management. The system significantly reduces manual reading effort, improves productivity, and enables users to quickly access important information from research papers, reports, articles, and other textual documents.

The responsive and intuitive user interface ensures smooth interaction for users across academic, research, professional, and organizational environments. The modular and scalable architecture of the system also allows future enhancements and integration of advanced AI capabilities without major structural changes.

Future enhancements may include:

- Integration of advanced deep learning and transformer models for more accurate abstractive summarization.
- Multilingual document summarization support for processing documents in different languages.
- Voice-based summary generation and text-to-speech functionality for accessibility improvements.
- Real-time collaborative document analysis and sharing features.
- Integration of plagiarism detection and semantic analysis functionalities.
- Mobile application development for Android and iOS platforms.
- Cloud-based deployment for handling large-scale enterprise-level document processing.
- AI-powered question-answering systems based on uploaded documents.

Overall, the AI-Powered Document Summarization System provides a reliable, scalable, and intelligent solution for modern document processing and information management requirements while establishing a strong foundation for future AI-driven innovations.

10.1.1 KEY ACHIEVEMENTS SUMMARY

The following table summarizes the key achievements of the AI-Powered Document Summarization System mapped to the original project objectives:

The AI-Powered Document Summarization System successfully achieved all major project objectives by providing an intelligent, secure, and efficient platform for automated document analysis and summarization. The system effectively generates concise summaries from lengthy documents using Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques, significantly reducing manual reading effort and improving productivity. It supports multiple document formats such as PDF, DOCX, and TXT, while also providing intelligent keyword extraction and important sentence identification. Secure JWT-based authentication and MongoDB-based data management ensure safe handling of user documents and summaries. Real-time processing updates using Socket.IO improve user experience during document upload and summarization processes. The scalable MERN stack architecture enables efficient handling of multiple users and large documents without major performance issues. Overall, the project successfully delivers a reliable, user-friendly, and AI-driven solution for modern document management and information extraction requirements while contributing to digital innovation and productivity enhancement.

10.1.2 FINAL CONCLUSION STATEMENT

In conclusion, the AI-Powered Document Summarization System successfully demonstrates how modern web technologies, Artificial Intelligence (AI), and Natural Language Processing (NLP) techniques can be utilized to solve real-world challenges related to document analysis and information management. The system provides a secure, scalable, efficient, and user-friendly platform for automatically generating meaningful summaries from lengthy textual documents.

The project effectively integrates React.js, Node.js, Express.js, MongoDB, and AI/NLP models to create a complete full-stack solution capable of document upload, text extraction, keyword identification, AI-based summarization, real-time processing updates, and secure document management. The application significantly reduces manual reading effort, improves productivity, and enhances information accessibility for students, researchers, professionals, and organizations.

The scalable MERN-based architecture, responsive user interface, and intelligent automation capabilities make the system suitable for academic, professional, and organizational environments requiring efficient document processing solutions. The project also contributes to digital transformation by promoting intelligent information management and automation using AI-driven technologies.

FUTURE WORK

While the AI-Powered Document Summarization System represents a comprehensive and intelligent solution in its current form, the following enhancements are planned for future development phases.

The scalable MERN-based architecture, responsive user interface, and intelligent automation capabilities make the system suitable for academic, professional, and organizational environments requiring efficient document processing solutions. The project also contributes to digital transformation by promoting intelligent information management and automation using AI-driven technologies.

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PROJECT SUMMARY

About Project

Title of the project	AI- POWERED DOCUMENT SUMMARIZATION SYSTEM
Semester	VIII Semester
Members	Deepak kurmi (0187CS221073), Ajay Ahirwar (0187CS221025), Alok Singh(0187CS221034), Anmol Bankar (0187CS221049)
Team Leader	Deepak Kurmi
Describe role of every member in the project	Deepak Kurmi: Overall architecture, Backend. Alok Singh: Frontend Ajay Ahirwar: UI/UX design and Documentation. Anmol Bankar: PPT Presentation
What is the motivation for selecting this project?	The motivation behind selecting this project is to develop an AI-based system that reduces manual reading effort by automatically generating concise summaries from lengthy documents.
Project Type (Desktop Application, Web Application, Mobile App, Web)	Web Application (AI-Based MERN Stack Application)

Tools & Technologies

Programming language used	Java & Python
Compiler used (with version)	VS Code (Visual Studio code)
IDE used (with version)	VS Code (Visual Studio code)
Front End Technologies (with version, wherever)	React.js

Applicable)	
Back End Technologies (with version, wherever applicable)	Java & JDK

Software Design & Coding

Is prototype of the software developed?	<u>Yes</u>
SDLC model followed	Agile Model
Why above SDLC model is followed?	Agile allows continuous development, testing, and improvement. It is suitable for mobile applications where features are added incrementally.
Justify that the SDLC model mentioned above is followed in the project.	The project was developed in phases such as authentication, visitor module, notification system, and emergency module, following iterative Agile development.
Software Design	Object-Oriented Approach
Name the diagrams developed (According to the Design approach followed)	• Use Case Diagram • ER Diagram • System Architecture Diagram
In case Object Oriented approach is followed, which of the OOPS principles are covered in	• Encapsulation • Abstraction • Inheritance • Polymorphism

design?	
Total no. of front-end pages	3-4 Screens (Login, Home)
Front end validations applied (Yes / No)	Yes
Session management done (in case of web applications)	Yes
Is application browser compatible (in case of web applications)	Yes Applicable
Exception handling done (Yes / No)	Yes
Commenting done in code (Yes / No)	Yes

Naming convention followed (Yes / No)	Yes (CamelCase, meaningful naming)
Total no. of Use-cases	1-2

Project Requirements

MVC architecture followed (Yes / No)	Yes
If yes, write the name of MVC architecture	MVM (Model ViewModel)

followed (MVC-1, MVC-2)	
Design Pattern used (Yes / No)	Yes
If yes, write the name of Design Pattern used	MVM Architecture Pattern
Interface type (CLI / GUI)	GUI (Graphical User Interface)
No. of Actors	2
Total no. of Functional Requirements	10-12
List few important non-Functional Requirements	• Security • Scalability • Performance • Reliability • Usability

Testing

Which test is performed? (Manual or Automation)	Manual Testing
Is Beta testing done for this project?	Yes

Write project narrative covering above mentioned points

The AI-Powered Document Summarization System is an intelligent web-based application developed to simplify document analysis and reduce the time required to read lengthy textual content. The system replaces manual document reading and information extraction processes with an automated AI-driven summarization solution that generates concise and meaningful summaries from large documents.

The application is developed using the MERN stack, including React.js for the frontend, Node.js and Express.js for backend services, and MongoDB for secure data management. Artificial Intelligence (AI) and Natural Language Processing (NLP) technologies such as transformer-based models, keyword extraction algorithms, and text preprocessing techniques are integrated to perform intelligent document summarization and analysis.

The development followed the Agile model, allowing incremental implementation of features such as user authentication, document upload, text extraction, AI-based summarization, keyword extraction, real-time processing updates, and summary management. The system provides users with a responsive and user-friendly interface for uploading PDF, DOCX, and TXT documents.

Deepak Kurmi

0187CS221073

Guide Signature

Ajay Ahirwar

0187CS221025

Dr. Shweta Singh

Alok Singh

0187CS221034

Anmol Bankar

0187CS221049

APPENDIX-1

GLOSSARY OF TERMS

(In alphabetical order)

A

AI (Artificial Intelligence)

Artificial Intelligence refers to computer systems capable of performing tasks that normally require human intelligence. In the AI-Powered Document Summarization System, AI is used to analyze documents and generate concise summaries automatically.

API (Application Programming Interface)

An API is a set of rules and protocols that allows communication between different software applications. REST APIs are used in the system for communication between frontend, backend, and AI modules.

D

Document Summarization

Document Summarization is the process of reducing lengthy textual content into a concise summary while preserving important information and context.

F

Frontend

Frontend refers to the user interface part of the application that users interact with directly. In this project, React.js is used to build the frontend interface.

N

NLP (Natural Language Processing)

Natural Language Processing is a branch of Artificial Intelligence that enables computers to understand, process, and analyze human language. In this project, NLP is used for text preprocessing, keyword extraction, and document summarization.

P

PDF

PDF (Portable Document Format) is a commonly used document format supported by the system for document upload and summarization.

R

React.js

React.js is a JavaScript library used for building responsive and interactive user interfaces in the frontend of the application.

REST API

REST API is an architectural style used for communication between frontend and backend services through HTTP requests

